

Algebra 1
Unit 14
Lesson 1 Homework

Name **Answer Key**
Date _____
Period _____

1. A survey of high school students asked participants if they were attending prom this year and whether they were an underclassman (freshman or sophomore) or an upperclassman (junior or senior). The data is summarized:
- 375 upperclassmen are planning to attend the prom
 - 189 underclassmen are not attending the prom
 - 403 participants are not attending the prom
 - There were 1000 participants in the survey

Complete the joint frequency table.

Student Prom Attendance

	Attending Prom	Not Attending Prom	Total
Underclassmen	222	189	411
Upperclassmen	375	214	589
Total	597	403	1000

2. How many underclassmen are not planning to attend the prom?

189

3. How many total students are planning to attend the prom?

597

4. How many underclassmen were surveyed?

411

5. Catalina public schools nutrition services department conducted a survey to find out students' favorite ice cream flavors. The data of this survey is shown in the table below.

Elementary Students		
Chocolate	Vanilla	Strawberry
230	441	195

Middle School Students		
Chocolate	Vanilla	Strawberry
530	526	406

High School Students		
Chocolate	Vanilla	Strawberry
944	905	471

Construct a two-way frequency table that shows all of the survey data.

Students

Ice Cream Flavors	Elementary	Middle School	High School	Total
Chocolate	230	530	944	1704
Vanilla	441	526	905	1872
Strawberry	195	406	471	1072
Total	836	1462	2320	4648

1. At Southside Middle School, 678 students were asked which of the school academies they would like to be in as their first choice. Sixty-four seventh graders said they would prefer the STEM academy. A total of 244 students said they would like to enroll in the Arts academy as their first choice, of which 88 are 7th graders. Of the 284 eighth graders, 79 of them would choose the STEM academy as their first choice. There are 194 sixth graders at Southside. Of the 216 Business academy choices, 78 are sixth graders.

Complete the two-way table summarizing the data.

Southside Middle School Academy Preference

	STEM	Arts	Business	Total
6 th	75	41	78	194
7 th	64	88	48	200
8 th	79	115	90	284
Total	218	244	216	678

2. A recent study was conducted among students to determine their most recent score based on the time, in hours, the participants spent studying. The following two-way table shows the resulting data.

Complete the table.

Test Score Survey

	0 – 50%	51 – 79%	80 – 100%	Total
0 – 2 hours	2	16	12	30
2 – 4 hours	1	10	8	19
4 + hours	0	0	36	36
Total	3	26	56	85

3. What is the smallest marginal frequency in the table?

3 students that scored 0 – 50% on the test

4. What is the largest joint frequency in the table?

36 students that studied for 4 + hours and scored 80 – 100% on the test

5. Describe what the value 10 means in the context of the data.

10 students that studied for 2 – 4 hours and scored 51 – 79% on the test

6. Describe what the value of 56 means in the context of the data.

56 students that scored 80 – 100% on the test

Algebra 1
Unit 14
Lesson 3 Homework

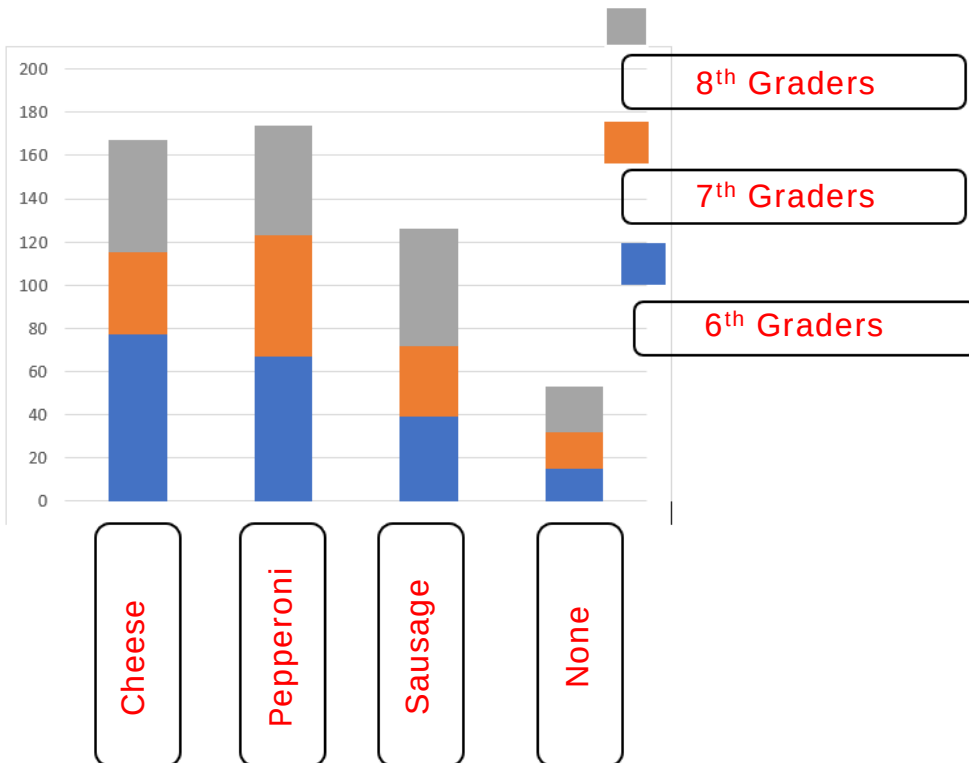
Name Answer Key
Date _____
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Westside Middle School conducted a school pizza party to celebrate the 100th day of school. Each student was given a choice of cheese, pepperoni, or sausage pizza. Some students did not want pizza. The results are shown in the frequency table below. A segmented bar graph was created with the data from the table. Use the figures to complete questions 1-5.

	6 th	7 th	8 th	Total
Cheese	77	38	52	167
Pepperoni	67	56	51	174
Sausage	39	33	54	126
None	15	17	21	53
Total	198	144	178	520

1. Circle the marginal frequencies in the table.

2. Box the joint frequencies in the table.



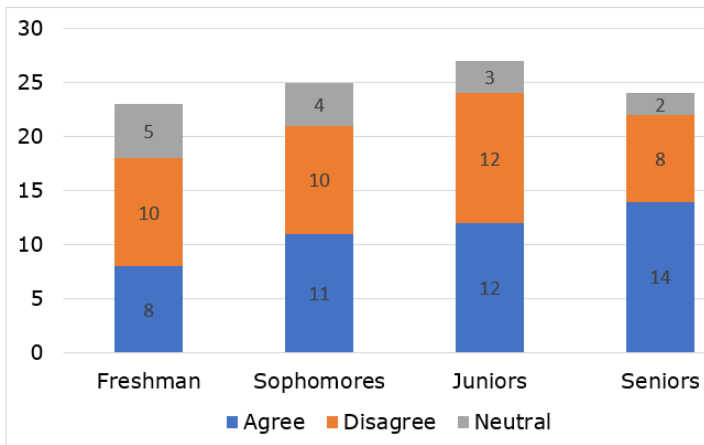
3. Label each marginal frequency in the boxes of the segmented bar graph.

4. Label the value for each joint frequency on the segmented bar graph.

5. Based on the data what type of pizza is the least favorite for each grade level?

6th graders' least favorite is sausage.
7th graders' least favorite is also sausage.
8th graders' least favorite is pepperoni.

The principal at a local high school is considering making uniforms mandatory to improve behavior. The principal polled members from the student council from each grade level to determine if students agree or disagree that uniforms will improve behavior. The results are included on the segmented bar graph below. Use the bar graph to complete questions 6-10.



6. What does the joint frequency value of 14 represent on the bar graph?

The number of seniors that agree that uniforms will improve behavior

7. How many Sophomores disagree that uniforms will improve behavior?

10

8. How many total students polled agree that uniforms will improve behavior?

$8 + 11 + 12 + 14 = 45$ students

9. How many Juniors and Seniors are neutral on the topic of uniforms?

3 Juniors and 2 Seniors

10. If the principal makes her decision based on the data in the segmented bar graph, should she make uniforms mandatory? Explain your reasoning.

Yes, because more students (45) agree than disagree (40).

Students were asked the following questions on a survey:

- A. Do you have no siblings, one sibling, or two or more siblings?
- B. Do you have dogs, cats, or no pets?

The two-way table gives partial results. Use this information to complete questions 1-5.

	No Siblings	One Sibling	Two or More Siblings	Total
Dogs	8	15	7	30
Cats	2	12	11	25
No Pets	1	3	6	10
Total	11	30	24	65

1. Complete the table assuming all students answered both questions.

2. What fraction of total students have one sibling and cats?

$$\frac{12}{65}$$

3. Does the fraction in #2 represent a marginal or joint frequency? Explain.

Joint frequency because it is found in the center and is a ratio of a part to the total.

4. What fraction of students with 2 or more siblings have dogs?

$$\frac{7}{24}$$

5. Does the fraction in #4 represent a marginal or joint frequency? Explain.

Joint frequency since it is the ratio of a part (7) to a total (24).

Coach Jenkins would like to know if there is any association between the height of the basketball players in his league and the height of their parents. He made a segmented bar graph from the information in the two-way table. Use this information to complete questions 6-10.

	Players under 6 feet	Players 6 feet & over	Total
Both Parents under 6 feet	34	38	72
Both Parents 6 feet & over	18	30	48
Total	52	68	120

6. What fraction of total basketball players surveyed have parents 6 feet and over?

$$\frac{48}{120} = \frac{2}{5}$$

7. What fraction of basketball players under 6 feet have both parents that are 6 feet and over?

$$\frac{18}{52} = \frac{9}{26}$$

8. What fraction of basketball players 6 feet and over have both parents that are 6 feet and over?

$$\frac{30}{68} = \frac{15}{34}$$

9. Explain whether the fractions in questions #6, 7, and 8 marginal or joint frequencies?

They are joint frequencies.

10. What possible associations can Coach Jenkins make about the height of players in his league to the height of their parents? **Answers may vary.**
There is a strong association between players that are under 6 ft and parents height, however there does not seem to be any association between players 6 ft and over.

Mrs. Washington, the school guidance counselor, at Ready High School is concerned about the growing number of students with jobs (part-time and/or full time). A school wide survey was conducted asking students about their job status. 1074 of Ready High School's 2000 students have some type of job.

Students in all grades were surveyed (9^{th} , 10^{th} , 11^{th} , and 12^{th}). The possible responses were Yes (part-time or full-time job) or No (no job).

- 23.5% of students surveyed were in the 9^{th} grade
- 27.6% of students surveyed were in the 12^{th} grade
- Of students without a job, 30.5% are 11^{th} grade students.
- Of students surveyed, 20.8% were in 10^{th} grade.
- 20.4% of 9^{th} grade students have a job.
- 45.4% of students with a job are in the 12^{th} grade

Create a relative two-way frequency table below for questions 1-4.
Round all answers to the nearest tenth of a percent.

	Part – Time / Full Time Job			
	Grade	Yes	No	Total
1.	9 th	4.8%	18.6%	23.5%
2.	10 th	10.5%	10.3%	20.8%
3.	11 th	14.0%	14.1%	28.1%
4.	12 th	24.4%	3.2%	27.6%
	Total	53.7%	46.3%	100%

Data was collected on the amount of time High School freshmen and seniors spend on social media each day. Students were surveyed and asked about 2 different intervals of daily social media time (0 – 4 hours OR Over 4 hours).

- 47.52% of students in the sample were freshmen.
- Of students that spend 0 – 4 hours each day on social media, 38.96% were seniors.
- 52.15% of students in the sample spent over 4 hours each day on social media.
- 56.28% of freshmen spend over 4 hours each day on social media.

Use this information to answer questions 5-8. Fill in the blank with either joint, marginal, or conditional.

5. 47.52% represents a marginal relative frequency.

6. 38.96% represents a conditional relative frequency.

7. 52.15% represents a marginal relative frequency.

8. 56.28% represents a conditional relative frequency.

9. Do you have enough information to create a two-way relative frequency chart from the information provided? Explain.

Yes, because a partial table can be made from the information given, and then the rest of the table can be filled in.

10. How can you distinguish between a joint relative frequency and a conditional relative frequency on a two-way frequency table? Explain.

Joint relative frequency is the ratio between a categorical value and the total, whereas conditional relative frequency is the ratio of joint relative frequency to marginal relative frequency.

In 2014 a study was conducted by a PhD candidate at Georgia Southern University examining the correlation between swimming ability, gender, and race. Data from that study included the following information in a two-way frequency table. Use this information to answer questions 1-6.

Ability	Male	Female	Total
Cannot Swim	16.07%	18.50%	34.57%
Can Swim a Little	19.14%	17.11%	36.25%
Can Swim Efficiently	15.37%	13.81%	29.18%
Total	50.58%	49.42%	100.00%

1. Interpret the joint relative frequency females who can swim a little.
17.11% of people surveyed were females who can swim a little.
2. Interpret the value of 34.57% in context of the data in the table.
34.57% of those surveyed cannot swim.
3. Interpret the value of 13.81% in context of the data in the table.
13.81% surveyed were females who can swim efficiently.
4. Interpret the value of $\frac{36.25\%}{100.00\%}$ in context of the data in the table.
Of those surveyed, people that can swim a little.
5. Interpret the value of $\frac{19.14\%}{50.58\%}$ in context of the data in the table.
The percentage of boys that can swim a little
6. What percentage of females can swim efficiently based on the data in the table?
 $\frac{13.81\%}{49.42\%} = 27.94\%$
7. What percentage of those who can swim efficiently are males?
 $\frac{15.37\%}{29.18\%} = 52.67\%$
8. Can you make any associations between gender and swimming ability based on the data? Explain.

No association can be made because the row conditional frequencies are all near 50% and all column conditional frequencies are near 33.3. %.

In the 2014 Georgia Southern University study the correlation between swimming ability and classification in school was also studied. Data from that study included the following information:

- 36.15% of all students were Sophomores.
- 24.34% of all students were Seniors.
- 36.27% of all students can swim a little.
- Of all students, 29.18% swim efficiently.
- 41.02% of seniors can swim a little.
- 7.91% of students that cannot swim are juniors.
- Of the students that cannot swim, 19.54% are sophomores.
- Of the seniors, 23.83% cannot swim.
- 23.75% of students that swim efficiently are juniors.
- Of those students that can swim efficiently, 19.72% are freshmen.
- 1.92% of students were freshmen that can swim a little.

Use this information to answer questions 9-10.

9. Based on the information above create a two-way relative frequency table. Round to the nearest hundredth.

Ability	Freshman	Sophomore	Junior	Senior	Total
Cannot Swim	19.27%	6.75%	2.73%	5.80%	34.55%
Can Swim a Little	1.92%	21.46%	2.91%	9.98%	36.27%
Can Swim Efficiently	5.75%	7.94%	6.93%	8.56%	29.18%
Total	26.94%	36.15%	12.57%	24.34%	100%

10. Does there appear to be an association between the abilities to swim and the class of students in school?

The relative frequencies in the table do appear to show a weak association between the grade a student is in and their ability to swim.